

TMUA REVISION RESOURCE

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r2drew2 edition

PAPER 1
MARK SCHEME

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Question 1

Because $x^2 + px + q$ has roots a and b ,

$$x^2 + px + q = (x - a)(x - b) = x^2 - (a + b)x + ab,$$

and comparing coefficients gives

$$a + b = -p, \quad ab = q.$$

The second quadratic $x^2 + qx + p$ has roots $a + 1$ and $b + 1$, so in the same way

$$(a + 1) + (b + 1) = -q, \quad (a + 1)(b + 1) = p.$$

Take the sum equation first. Its left-hand side is $a + b + 2$, and $a + b = -p$, so

$$-p + 2 = -q \quad \implies \quad q = p - 2. \tag{1}$$

Now the product equation. Expanding,

$$(a + 1)(b + 1) = ab + a + b + 1 = q + (-p) + 1 = q - p + 1,$$

and this equals p , so

$$q - p + 1 = p. \tag{2}$$

Substituting (1) into (2):

$$(p - 2) - p + 1 = p \quad \implies \quad -1 = p.$$

Then $q = p - 2 = -3$, and

$$p + q = -1 + (-3) = \boxed{-4}.$$

Answer: option D, -4 .

Question 2

In $(1+x)^n(1-x)$, the coefficient of x^2 comes from x^2 in $(1+x)^n$ multiplied by 1, and from x^1 in $(1+x)^n$ multiplied by $-x$. So it is

$$\binom{n}{2} - \binom{n}{1} = \frac{n(n-1)}{2} - n = \frac{n(n-3)}{2}.$$

Likewise the coefficient of x^3 is

$$\binom{n}{3} - \binom{n}{2} = \frac{n(n-1)(n-2)}{6} - \frac{n(n-1)}{2} = \frac{n(n-1)[(n-2)-3]}{6} = \frac{n(n-1)(n-5)}{6}.$$

Setting them equal:

$$\frac{n(n-3)}{2} = \frac{n(n-1)(n-5)}{6}.$$

Since $n \geq 3$ we have $n \neq 0$, so divide by n and multiply through by 6:

$$3(n-3) = (n-1)(n-5).$$

Expanding,

$$3n - 9 = n^2 - 6n + 5 \implies n^2 - 9n + 14 = 0 \implies (n-2)(n-7) = 0.$$

The condition $n \geq 3$ rules out $n = 2$, so

$$\boxed{n = 7}.$$

Answer: option D, $n = 7$.

Question 3

Divide the whole equation by 4^x , which is never zero:

$$\left(\frac{9}{4}\right)^x - 10\left(\frac{3}{2}\right)^x + 9 = 0.$$

Now $\frac{9}{4} = \left(\frac{3}{2}\right)^2$, so $\left(\frac{9}{4}\right)^x = \left(\left(\frac{3}{2}\right)^x\right)^2$. Setting

$$u = \left(\frac{3}{2}\right)^x,$$

the equation becomes a quadratic:

$$u^2 - 10u + 9 = 0 \implies (u - 1)(u - 9) = 0.$$

So $u = 1$ or $u = 9$. Since $u = (3/2)^x$ is a one-to-one function of x , each value of u gives exactly one solution, and the two solutions p and q satisfy

$$\left(\frac{3}{2}\right)^p = 1, \quad \left(\frac{3}{2}\right)^q = 9$$

in some order.

Here is the point of the question: you never need to find p or q . By the index law,

$$\left(\frac{3}{2}\right)^{p+q} = \left(\frac{3}{2}\right)^p \left(\frac{3}{2}\right)^q = 1 \times 9 = \boxed{9}.$$

Answer: option 2, 9.

Question 4

Completing the square,

$$f(x) = x^2 - 6x + 7 = (x - 3)^2 - 2,$$

so the graph is symmetric about $x = 3$.

The first condition. $f(a) = f(b)$ gives $(a - 3)^2 = (b - 3)^2$, so

$$a - 3 = b - 3 \quad \text{or} \quad a - 3 = -(b - 3).$$

The first would force $a = b$, which is excluded because a and b are distinct. Hence

$$a + b = 6. \tag{1}$$

The second condition. $f(a+1) = f(b-2)$ gives $(a+1-3)^2 = (b-2-3)^2$, that is $(a-2)^2 = (b-5)^2$, so

$$a - 2 = b - 5 \quad \text{or} \quad a - 2 = -(b - 5).$$

Both branches must now be tested against (1).

The second branch gives $a - 2 = -b + 5$, so $a + b = 7$. This contradicts (1), so it is impossible.

The first branch gives

$$a - b = -3. \tag{2}$$

Solving. Adding (1) and (2): $2a = 3$, so $a = \frac{3}{2}$ and $b = \frac{9}{2}$. Therefore

$$ab = \frac{3}{2} \cdot \frac{9}{2} = \boxed{\frac{27}{4}}.$$

Answer: option r, $\frac{27}{4}$.

Question 5

Factorising,

$$n^2 - 9 = (n - 3)(n + 3) = 2^k.$$

Since $n > 3$, both factors are positive integers. If either factor had an odd prime divisor, that prime would divide 2^k , which is impossible. So *both factors are themselves powers of 2*.

Their difference is fixed:

$$(n + 3) - (n - 3) = 6.$$

So we need two powers of 2 differing by 6. Listing them — 1, 2, 4, 8, 16, 32, ... — the gaps between consecutive powers are 1, 2, 4, 8, 16, ..., and once both powers are 8 or larger the difference is at least 8. So only the small cases can work, and checking them:

$$2 - 1 = 1, \quad 4 - 1 = 3, \quad 4 - 2 = 2, \quad 8 - 1 = 7, \quad 8 - 2 = 6, \quad \checkmark \quad 8 - 4 = 4, \quad 16 - 8 = 8.$$

The only pair is 2 and 8. Hence

$$n - 3 = 2, \quad n + 3 = 8,$$

both giving $n = 5$. Then

$$2^k = 5^2 - 9 = 16 = 2^4,$$

so

$$\boxed{k = 4}.$$

Answer: option 2, $k = 4$.

Question 6

Any point on both circles satisfies both equations, so it satisfies their difference:

$$(x^2 + y^2 - 25) - ((x - 8)^2 + y^2 - r^2) = 0.$$

The y^2 terms cancel, and $x^2 - (x - 8)^2 = 16x - 64$, so

$$16x - 64 - 25 + r^2 = 0 \implies 16x = 89 - r^2.$$

This involves x alone, so the common chord lies on a *vertical* line.

Now use the first circle, which has centre the origin and radius 5. The chord has length 8, so half of it is 4. If d is the distance from the origin to the chord, the right-angled triangle formed by the centre, the midpoint of the chord and one endpoint gives

$$d^2 + 4^2 = 5^2 \implies d = 3.$$

A vertical line at distance 3 from the origin is $x = 3$ or $x = -3$, and this is where the two possible values of r^2 come from.

The endpoints of the chord have $y = \pm 4$ in each case, since $3^2 + 4^2 = 25$.

If $x = 3$: take the endpoint $(3, 4)$. It lies on the second circle, so

$$r^2 = (3 - 8)^2 + 4^2 = 25 + 16 = 41.$$

If $x = -3$: take the endpoint $(-3, 4)$. Then

$$r^2 = (-3 - 8)^2 + 4^2 = 121 + 16 = 137.$$

The sum is

$$41 + 137 = \boxed{178}.$$

Answer: option D, 178.

Question 7

The given information says that for some polynomial Q ,

$$P(x) = (x - 1)^2 Q(x) + 4x - 1.$$

This is an identity, so we may replace x throughout by anything. Write $g(x) = x^2 - 3x + 3$ and substitute:

$$P(g(x)) = (g(x) - 1)^2 Q(g(x)) + 4g(x) - 1.$$

Now examine $g(x) - 1$:

$$g(x) - 1 = x^2 - 3x + 2 = (x - 1)(x - 2).$$

Therefore

$$(g(x) - 1)^2 = (x - 1)^2 (x - 2)^2,$$

which contains a factor of $(x - 2)^2$. So the entire first term is divisible by $(x - 2)^2$ and contributes nothing to the remainder.

All that remains is to divide

$$4g(x) - 1 = 4(x^2 - 3x + 3) - 1 = 4x^2 - 12x + 11$$

by $(x - 2)^2 = x^2 - 4x + 4$. Since

$$4(x^2 - 4x + 4) = 4x^2 - 16x + 16,$$

subtracting gives

$$(4x^2 - 12x + 11) - (4x^2 - 16x + 16) = 4x - 5.$$

The part subtracted is a multiple of $(x - 2)^2$, so the remainder is

$$\boxed{4x - 5}.$$

Answer: option E, $4x - 5$.

Question 8

Let $u = x^2 - 5x$. Then $x^2 - 5x + 4 = u + 4$, so

$$F(x) = |u| + |u + 4|.$$

Minimising $|u| + |u + 4|$. Treat the three cases.

If $u > 0$, then $|u| = u$ and $|u + 4| = u + 4$, so $F = 2u + 4 > 4$.

If $-4 \leq u \leq 0$, then $|u| = -u$ and $|u + 4| = u + 4$, so

$$F = -u + u + 4 = 4.$$

If $u < -4$, then $|u| = -u$ and $|u + 4| = -(u + 4)$, so $F = -2u - 4 > 4$.

So the least value of F is 4, attained exactly when $-4 \leq u \leq 0$.

Translating back. We need

$$-4 \leq x^2 - 5x \leq 0.$$

The right inequality: $x^2 - 5x \leq 0$ means $x(x - 5) \leq 0$, so

$$0 \leq x \leq 5.$$

The left inequality: $x^2 - 5x \geq -4$ means $x^2 - 5x + 4 \geq 0$, that is $(x - 1)(x - 4) \geq 0$, so

$$x \leq 1 \quad \text{or} \quad x \geq 4.$$

Intersecting the two:

$$0 \leq x \leq 1 \quad \text{or} \quad 4 \leq x \leq 5.$$

Both intervals are closed, matching the statement in the question. Their lengths are $1 - 0 = 1$ and $5 - 4 = 1$, so the sum is

$$1 + 1 = \boxed{2}.$$

Answer: option 2, 2.

Question 9

Label $BC = t$. The angle at B is between the sides AB and BC , and the side opposite it is AC . The cosine rule gives

$$AC^2 = AB^2 + BC^2 - 2 \cdot AB \cdot BC \cos(\angle ABC),$$

that is

$$7^2 = 12^2 + t^2 - 2(12)t \cos 30^\circ.$$

Since $\cos 30^\circ = \frac{\sqrt{3}}{2}$, the last term is $12\sqrt{3}t$, so

$$49 = 144 + t^2 - 12\sqrt{3}t \implies t^2 - 12\sqrt{3}t + 95 = 0. \quad (*)$$

Why there are two triangles. The discriminant of $(*)$ is

$$(12\sqrt{3})^2 - 4(95) = 432 - 380 = 52 > 0,$$

so there are two distinct real roots; and since their sum $12\sqrt{3}$ and product 95 are both positive, both roots are positive. Two valid triangles, as stated.

The key step. Let the two roots be t_1 and t_2 . Then

$$t^2 - 12\sqrt{3}t + 95 = (t - t_1)(t - t_2) = t^2 - (t_1 + t_2)t + t_1t_2,$$

so comparing constant terms,

$$t_1t_2 = 95.$$

The areas. For either triangle, the area is

$$K = \frac{1}{2} \cdot AB \cdot BC \cdot \sin(\angle ABC) = \frac{1}{2}(12)t \sin 30^\circ = \frac{1}{2}(12)t \cdot \frac{1}{2} = 3t.$$

Therefore

$$K_1K_2 = (3t_1)(3t_2) = 9t_1t_2 = 9(95) = \boxed{855}.$$

Note that the two values of t are never needed individually. They involve $\sqrt{52}$ and would make the arithmetic far worse.

Answer: option D, 855.

Question 10

Let $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$. From the given ratios,

$$\overrightarrow{OP} = \frac{2}{5}\mathbf{a}, \quad \overrightarrow{OQ} = \frac{3}{7}\mathbf{b}.$$

R on AQ . Travelling a fraction t of the way from A to Q ,

$$\overrightarrow{OR} = \mathbf{a} + t\left(\frac{3}{7}\mathbf{b} - \mathbf{a}\right) = (1-t)\mathbf{a} + \frac{3t}{7}\mathbf{b}.$$

R on BP . Travelling a fraction s of the way from B to P ,

$$\overrightarrow{OR} = \mathbf{b} + s\left(\frac{2}{5}\mathbf{a} - \mathbf{b}\right) = \frac{2s}{5}\mathbf{a} + (1-s)\mathbf{b}.$$

Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel, the coefficients must match separately:

$$1-t = \frac{2s}{5}, \quad \frac{3t}{7} = 1-s.$$

From the second, $s = 1 - \frac{3t}{7}$. Substituting into the first:

$$1-t = \frac{2}{5}\left(1 - \frac{3t}{7}\right).$$

Multiplying by 35:

$$35 - 35t = 14 - 6t \implies 21 = 29t \implies t = \frac{21}{29}.$$

Hence

$$\overrightarrow{OR} = \left(1 - \frac{21}{29}\right)\mathbf{a} + \frac{3}{7} \cdot \frac{21}{29}\mathbf{b} = \frac{8}{29}\mathbf{a} + \frac{9}{29}\mathbf{b}.$$

Finding S . The point S lies on the line OR , so $\overrightarrow{OS} = \lambda(8\mathbf{a} + 9\mathbf{b})$ for some λ . It also lies on AB , and a point $\alpha\mathbf{a} + \beta\mathbf{b}$ lies on the line AB exactly when $\alpha + \beta = 1$. So $17\lambda = 1$, giving

$$\overrightarrow{OS} = \frac{8}{17}\mathbf{a} + \frac{9}{17}\mathbf{b}.$$

Comparing with the general point $(1-\mu)\mathbf{a} + \mu\mathbf{b}$, which is a fraction μ of the way from A to B , we read off $\mu = \frac{9}{17}$. Therefore

$$AS : SB = \frac{9}{17} : \frac{8}{17} = \boxed{9 : 8}.$$

Answer: option 2, 9 : 8.

Question 11

Let

$$u = x^2 - 5x + 8.$$

The second expression is

$$8 - x^2 + 5x = 16 - (x^2 - 5x + 8) = 16 - u.$$

So the two arguments always sum to 16, whatever x is.

Combining the logarithms,

$$\log_2(u(16 - u)) = 4 \implies u(16 - u) = 2^4 = 16.$$

Rearranging,

$$u^2 - 16u + 16 = 0 \implies u = \frac{16 \pm \sqrt{256 - 64}}{2} = 8 \pm 4\sqrt{3}.$$

Now each value of u has to be converted back into an equation in x , and each must be tested.

Case $u = 8 - 4\sqrt{3}$. Then $x^2 - 5x + 8 = 8 - 4\sqrt{3}$, so

$$x^2 - 5x + 4\sqrt{3} = 0.$$

Its discriminant is $25 - 16\sqrt{3}$. Since $\sqrt{3} > 1.7$, we have $16\sqrt{3} > 27 > 25$, so the discriminant is negative and there are *no real solutions* from this case.

Case $u = 8 + 4\sqrt{3}$. Then

$$x^2 - 5x - 4\sqrt{3} = 0,$$

with discriminant $25 + 16\sqrt{3} > 0$, giving two distinct real solutions. These are the p and q in the question.

Both logarithms are defined here: $u = 8 + 4\sqrt{3} \approx 14.93 > 0$ and $16 - u \approx 1.07 > 0$.

Finally, comparing $x^2 - 5x - 4\sqrt{3}$ with $(x - p)(x - q) = x^2 - (p + q)x + pq$, the constant terms give

$$pq = \boxed{-4\sqrt{3}}.$$

Answer: option 2, $-4\sqrt{3}$.

Question 12

Write $t = x^{3/2}$. Then

$$x^3 = t^2, \quad x^6 = t^4,$$

so the fraction becomes

$$\frac{t^4 + t^2 + 1}{t^2 + t + 1}.$$

Now factorise the numerator. The trick is to complete a square:

$$t^4 + t^2 + 1 = (t^4 + 2t^2 + 1) - t^2 = (t^2 + 1)^2 - t^2,$$

which is a difference of two squares:

$$(t^2 + 1)^2 - t^2 = (t^2 + 1 - t)(t^2 + 1 + t) = (t^2 - t + 1)(t^2 + t + 1).$$

The factor $t^2 + t + 1$ cancels with the denominator, leaving $t^2 - t + 1$. Returning to x ,

$$\frac{x^6 + x^3 + 1}{x^3 + x^{3/2} + 1} = x^3 - x^{3/2} + 1.$$

(The denominator $x^3 + x^{3/2} + 1$ is never zero for $x \geq 0$, since all three terms are non-negative and the last is 1, so the cancellation is valid throughout $[0, 1]$.)

Now integrate term by term:

$$\int_0^1 (x^3 - x^{3/2} + 1) dx = \left[\frac{x^4}{4} - \frac{x^{5/2}}{5/2} + x \right]_0^1 = \left[\frac{x^4}{4} - \frac{2x^{5/2}}{5} + x \right]_0^1.$$

At $x = 1$ this is $\frac{1}{4} - \frac{2}{5} + 1$, and at $x = 0$ it is 0. So the integral is

$$\frac{1}{4} - \frac{2}{5} + 1 = \frac{5 - 8 + 20}{20} = \boxed{\frac{17}{20}}.$$

Answer: option R, $\frac{17}{20}$.

Question 13

Write

$$N = x^4 + 4x^2 + 1, \quad D = x^4 + 2x^3 + 3x^2 - 2x + 1,$$

so $F(x) = N/D$.

Step 1: the denominator is positive. Observe that

$$x^2(x+1)^2 + 2\left(x - \frac{1}{2}\right)^2 + \frac{1}{2} = (x^4 + 2x^3 + x^2) + (2x^2 - 2x + \frac{1}{2}) + \frac{1}{2} = x^4 + 2x^3 + 3x^2 - 2x + 1 = D.$$

Every term on the left is non-negative and the last is strictly positive, so $D > 0$ for all x . This matters: it means we may multiply an inequality by D without reversing it.

Step 2: test the value $\frac{3}{4}$. We want to know whether $\frac{N}{D} \geq \frac{3}{4}$ always holds, which (since $D > 0$) is equivalent to

$$4N - 3D \geq 0.$$

Computing,

$$4N = 4x^4 + 16x^2 + 4, \quad 3D = 3x^4 + 6x^3 + 9x^2 - 6x + 3,$$

so

$$4N - 3D = x^4 - 6x^3 + 7x^2 + 6x + 1.$$

This factorises as a perfect square:

$$x^4 - 6x^3 + 7x^2 + 6x + 1 = (x^2 - 3x - 1)^2.$$

Since a square is never negative, $4N - 3D \geq 0$, and therefore

$$F(x) \geq \frac{3}{4} \quad \text{for all } x > 0.$$

Step 3: the bound is attained. Equality needs $x^2 - 3x - 1 = 0$, that is

$$x = \frac{3 \pm \sqrt{13}}{2}.$$

The root $x = \frac{3+\sqrt{13}}{2} \approx 3.303$ is positive, so it lies in the domain and the value $\frac{3}{4}$ is actually reached.

Hence the least value is

$$\boxed{\frac{3}{4}}.$$

Answer: option E, $\frac{3}{4}$.

Question 14

Let S be the area of triangle ABC .

Setting up the three fractions. Let u be the perpendicular distance from P to BC , divided by the full height from A to BC . Define v and w in the same way with respect to CA and AB .

The three triangles PBC , PCA and PAB together fill ABC exactly. Triangle PBC has the same base BC as ABC but height u times as large, so its area is uS ; similarly for the others. Therefore

$$uS + vS + wS = S \implies u + v + w = 1.$$

The scale factors. The line through P parallel to BC cuts off a triangle at A . That triangle is similar to ABC , and its height from A is the full height minus the distance from P to BC — that is, a fraction $1 - u$ of the whole. So its length scale factor is $1 - u$. Likewise the triangles at B and C have scale factors $1 - v$ and $1 - w$.

Adding these:

$$(1 - u) + (1 - v) + (1 - w) = 3 - (u + v + w) = 3 - 1 = 2. \quad (*)$$

This is the heart of the question: *the three scale factors always sum to 2*, wherever P is placed.

Using the areas. Areas scale as the square of lengths, so the triangle at A with area 4 has scale factor $\sqrt{4/S} = 2/\sqrt{S}$, and similarly the others give $3/\sqrt{S}$ and $4/\sqrt{S}$. Substituting into (*):

$$\frac{2}{\sqrt{S}} + \frac{3}{\sqrt{S}} + \frac{4}{\sqrt{S}} = 2 \implies \frac{9}{\sqrt{S}} = 2 \implies \sqrt{S} = \frac{9}{2}.$$

Therefore

$$S = \left(\frac{9}{2}\right)^2 = \boxed{\frac{81}{4}}.$$

Answer: option 2, $\frac{81}{4}$.

Question 15

Squaring both sides of $|\sin \theta| + |\cos \theta| = \frac{1 + \sqrt{3}}{2}$:

$$\sin^2 \theta + 2|\sin \theta \cos \theta| + \cos^2 \theta = \left(\frac{1 + \sqrt{3}}{2}\right)^2 = 1 + \frac{\sqrt{3}}{2}.$$

Using $\sin^2 \theta + \cos^2 \theta = 1$ on the left,

$$|\sin \theta \cos \theta| = \frac{\sqrt{3}}{4}.$$

Now write $a = |\sin \theta|$ and $b = |\cos \theta|$. We know

$$a + b = \frac{1 + \sqrt{3}}{2}, \quad ab = \frac{\sqrt{3}}{4},$$

so a and b are the two roots of

$$4t^2 - 2(1 + \sqrt{3})t + \sqrt{3} = 0.$$

Solving gives

$$t = \frac{\sqrt{3}}{2} \quad \text{or} \quad \frac{1}{2}.$$

So $\{|\sin \theta|, |\cos \theta|\} = \left\{\frac{1}{2}, \frac{\sqrt{3}}{2}\right\}$: the reference angle is $\frac{\pi}{6}$ or $\frac{\pi}{3}$.

Each of the four quadrants of $[0, 2\pi)$ contains one angle with reference angle $\frac{\pi}{6}$ and one with $\frac{\pi}{3}$, giving eight solutions:

$$\frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{11\pi}{6}.$$

Pairing each angle with $2\pi - \theta$, there are four pairs each summing to 2π . Hence the total is

$$4(2\pi) = \boxed{8\pi}.$$

Answer: option D, 8π .

Question 16

Let $P = (h, k)$, and let the x -coordinates of B and C be u and v .

The normal at a general point. A point on the parabola is (t, t^2) . Since $\frac{dy}{dx} = 2x$, the tangent gradient there is $2t$, so the normal has gradient $-\frac{1}{2t}$ and equation

$$y - t^2 = -\frac{1}{2t}(x - t).$$

Multiplying by $2t$ and rearranging,

$$x + 2ty = 2t^3 + t.$$

This form is valid at $t = 0$ as well, where it gives the vertical normal $x = 0$.

Imposing that P lies on it. Substituting (h, k) ,

$$h + 2tk = 2t^3 + t \implies 2t^3 + (1 - 2k)t - h = 0. \quad (*)$$

The three parameters $1, u, v$ are exactly the roots of $(*)$, so

$$2(t - 1)(t - u)(t - v) = 2t^3 + (1 - 2k)t - h.$$

Comparing coefficients. Expanding the left-hand side,

$$2(t - 1)(t - u)(t - v) = 2t^3 - 2(1 + u + v)t^2 + 2(u + v + uv)t - 2uv.$$

The right-hand side of $(*)$ has no t^2 term, so

$$-2(1 + u + v) = 0 \implies u + v = -1. \quad (1)$$

Comparing constant terms,

$$-2uv = -h \implies h = 2uv. \quad (2)$$

Using the y -sum. The y -coordinates are $1, u^2$ and v^2 , and they sum to 14:

$$1 + u^2 + v^2 = 14 \implies u^2 + v^2 = 13.$$

Now

$$(u + v)^2 = u^2 + 2uv + v^2,$$

so using (1),

$$1 = 13 + 2uv \implies uv = -6.$$

Finally, from (2),

$$h = 2(-6) = \boxed{-12}.$$

Answer: option 2, -12 .

Question 17

Let the gradient be m . Since the line passes through $(1, -9)$,

$$y + 9 = m(x - 1), \quad \text{i.e.} \quad y = mx - m - 9.$$

Where it meets the curve,

$$x^3 - 12x^2 + 7x = mx - m - 9,$$

so the three intersection x -coordinates are the roots of

$$x^3 - 12x^2 + (7 - m)x + (m + 9) = 0. \quad (*)$$

Naming the roots. They form a geometric progression, so write them as

$$\frac{t}{r}, \quad t, \quad tr,$$

where t is the middle term and r the common ratio.

Multiplying the outer two factors first and writing $s = \frac{1}{r} + r$,

$$\left(x - \frac{t}{r}\right)(x - tr) = x^2 - tsx + t^2.$$

Multiplying by $(x - t)$ gives

$$x^3 - t(s + 1)x^2 + t^2(s + 1)x - t^3.$$

Comparing coefficients with $(*)$:

$$t(s + 1) = 12, \quad t^2(s + 1) = 7 - m, \quad -t^3 = m + 9.$$

Dividing the second equation by the first eliminates s :

$$t = \frac{7 - m}{12} \implies m = 7 - 12t.$$

Substitute this into $-t^3 = m + 9$:

$$-t^3 = 16 - 12t \implies t^3 - 12t + 16 = 0.$$

Factorising,

$$t^3 - 12t + 16 = (t - 2)^2(t + 4).$$

So $t = 2$ or $t = -4$. The x -coordinates must be positive, so $t = 2$. Hence

$$m = 7 - 12(2) = \boxed{-17}.$$

Answer: option D, -17 .

Question 18

At an intersection,

$$|x^2 - 1| = |(x - a)^2 - 1|,$$

which happens exactly when the two expressions are equal or are negatives of each other.

Possibility 1: equal.

$$x^2 - 1 = (x - a)^2 - 1 \implies a(a - 2x) = 0.$$

If $a = 0$ the two graphs are identical and there are infinitely many intersection points. So $a \neq 0$, and then

$$x = \frac{a}{2},$$

exactly one point.

Possibility 2: negatives.

$$x^2 - 1 = -((x - a)^2 - 1)$$

rearranges to

$$2x^2 - 2ax + a^2 - 2 = 0. \tag{*}$$

Its discriminant is

$$(-2a)^2 - 4(2)(a^2 - 2) = 16 - 4a^2.$$

So (*) has two distinct real roots exactly when

$$-2 < a < 2,$$

one repeated root when $a = \pm 2$, and none when $|a| > 2$.

Counting. Substituting $x = a/2$ into (*) gives $a^2/2 - 2$, which is zero exactly when $a = \pm 2$. So for $-2 < a < 2$ with $a \neq 0$, the point from Possibility 1 is distinct from the two from Possibility 2, giving three points.

Therefore exactly three intersections occur when

$$\boxed{-2 < a < 0 \text{ or } 0 < a < 2}.$$

Answer: option 2, $-2 < a < 0$ or $0 < a < 2$.

Question 19

Place the rectangle with

$$A = (0, 0), \quad B = (w, 0), \quad C = (w, h), \quad D = (0, h),$$

so that $wh = 3$. The path runs $A \rightarrow P \rightarrow Q \rightarrow R \rightarrow B$, meeting BC , then CD , then DA .

Unfolding. Reflect the destination B successively in the mirrors, working backwards:

$$B = (w, 0) \longrightarrow (-w, 0) \longrightarrow (-w, 2h) \longrightarrow (3w, 2h).$$

Reflection preserves distances, so the broken path has the same total length as an unbroken path from $(0, 0)$ to $(3w, 2h)$. Therefore

$$AP + PQ + QR + RB \geq \sqrt{9w^2 + 4h^2}.$$

Minimising over the rectangle. Using $wh = 3$,

$$9w^2 + 4h^2 = (3w - 2h)^2 + 12wh = (3w - 2h)^2 + 36 \geq 36.$$

Hence

$$AP + PQ + QR + RB \geq 6.$$

Equality. We need $3w = 2h$. Combined with $wh = 3$, this gives

$$w = \sqrt{2}, \quad h = \frac{3}{\sqrt{2}}.$$

The straight line in the unfolded picture folds back to points on the required sides, so the bound is attained. Thus the least possible value is

$$\boxed{6}.$$

Answer: option r, 6.

Question 20

Let S_n be the set of real solutions of $f_n(x) = 2$, and let $N_n = |S_n|$.

Base case. $f_1(x) = 2$ means $x^2 - 2 = 2$, so $x = \pm 2$ and

$$S_1 = \{2, -2\}, \quad N_1 = 2.$$

Stepping backwards. To solve $f(x) = c$, that is $x^2 - 2 = c$, we need $x^2 = c + 2$. So the number of solutions is

$$\begin{cases} 2 & \text{if } c > -2, \\ 1 & \text{if } c = -2, \\ 0 & \text{if } c < -2. \end{cases}$$

Every element of every S_n lies in $[-2, 2]$, and exactly one of them equals -2 . Indeed, this is true for S_1 , and taking preimages of a value in $[-2, 2]$ again gives values in $[-2, 2]$. Also -2 remains a solution at every stage.

So when forming S_{n+1} , every element of S_n contributes two new values except -2 , which contributes only one:

$$N_{n+1} = 2N_n - 1.$$

Iterating.

$$N_1 = 2, \quad N_2 = 3, \quad N_3 = 5, \quad N_4 = 9, \quad N_5 = 17, \quad N_6 = 33, \quad N_7 = 65, \quad N_8 = \boxed{129}.$$

Equivalently, $N_{n+1} - 1 = 2(N_n - 1)$, so

$$N_n = 2^{n-1} + 1.$$

Answer: option r, 129.

End of mark scheme